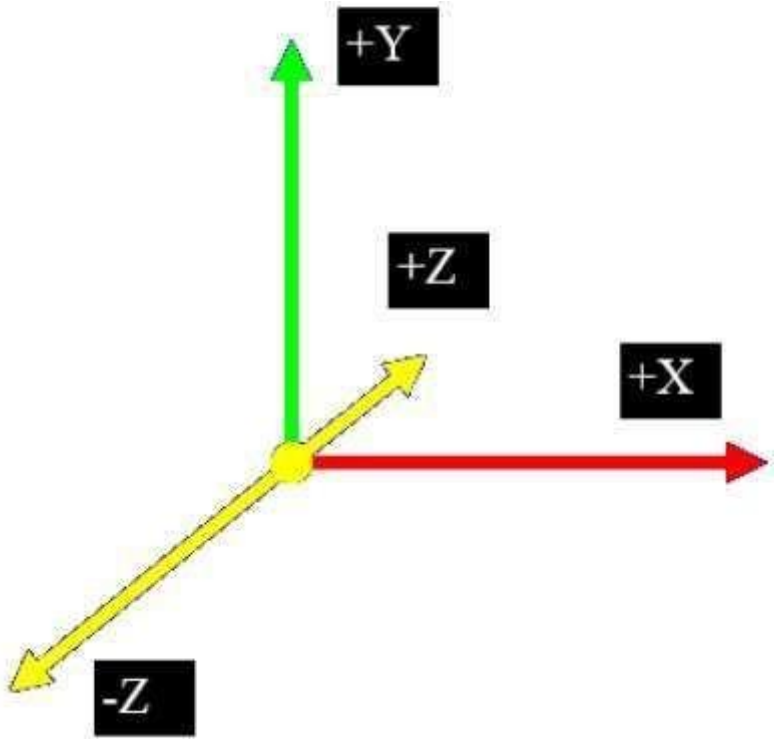
J 3D coordinate systems

In the 2D system, we use only two coordinates X and Y but in 3D, an extra coordinate Z is added. 3D graphics techniques and their application are fundamental to the entertainment, games, and computer-aided design industries. It is a continuing area of research in scientific visualization.

Furthermore, 3D graphics components are now a part of almost every personal computer and, although traditionally intended for graphics-intensive software such as games, they are increasingly being used by other applications.

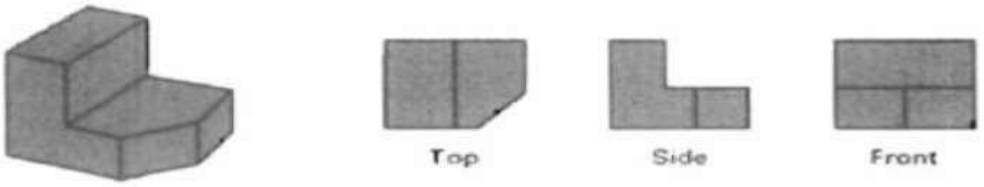


J 3-D display methods:

- Parallel projection,
- perspective projection

❖ **Parallel Projection:**

- Parallel projection is a method for generating a view of a solid object is to project points on the object surface along parallel lines onto the display plane.
- In parallel projection, parallel lines in the world coordinate scene project into parallel lines on the two-dimensional display planes.
- This technique is used in engineering and architectural drawings to represent an object with a set of views that maintain relative proportions of the object.
- The appearance of the solid object can be reconstructed from the major views



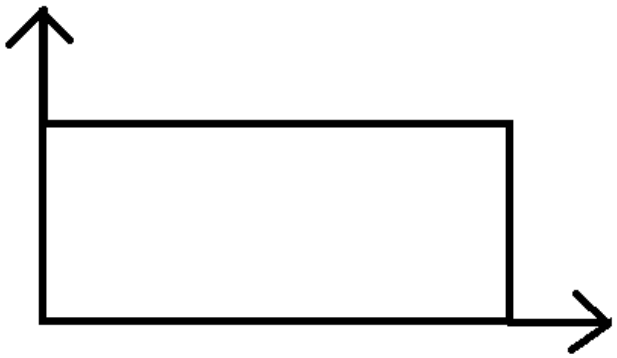
❖ **Perspective Projection:**

- It is a method for generating a view of a three-dimensional scene is to project points to the display plane alone converging paths.
- This makes objects further from the viewing position be displayed smaller than objects of the same size that are nearer to the viewing position.
- In a perspective projection, parallel lines in a scene that are not parallel to the display plane are projected into converging lines.

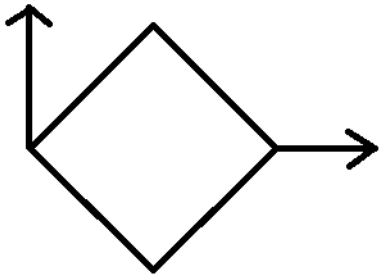
- The following figure shows all the three types of perspective projection –

Different types of perspective projections:

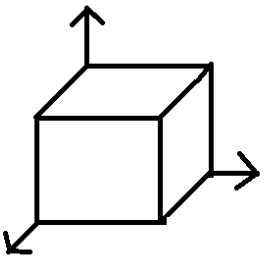
One point perspective projections: In this, principal axis has a finite vanishing point. Perspective projection is simple to draw.



Two point perspective projections: Exactly 2 principals have vanishing points. Perspective projection gives better impression of depth.



Three point perspective projections: All the three principal axes have finite vanishing point. Perspective projection is most difficult to draw.



❖ Introduction of 3D Object Representations

Representation schemes for solid objects are divided into two categories as follows:

1. Boundary Representation (B-reps)

- It describes a three-dimensional object as a set of surfaces that separate the object interior from the environment.
- Examples are polygon facets and spline patches.

2. Space Partitioning representation

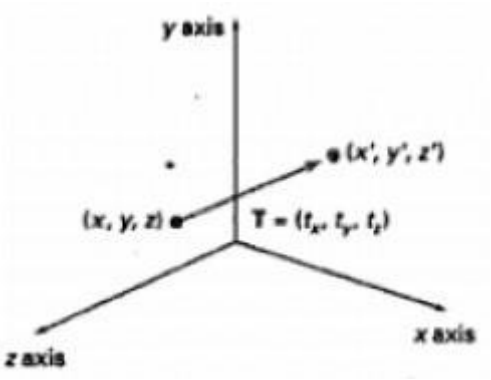
- Eg: Octree Representation

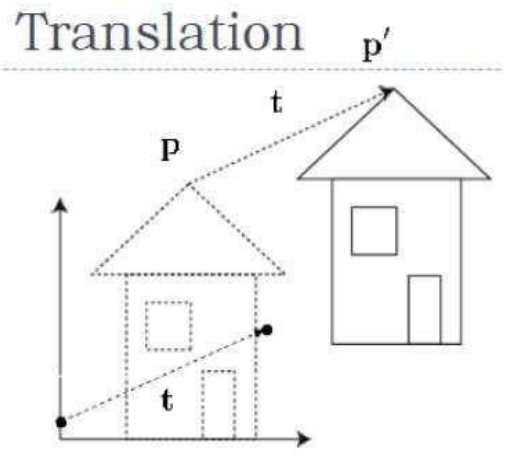
3D transformations:

❖ Translation

In 3D translation, we transfer the Z coordinate along with the X and Y coordinates. The process for translation in 3D is similar to 2D translation. A translation moves an object into a different position on the screen.

The following figure shows the effect of translation −





A point can be translated in 3D by adding translation coordinate (tx,ty,tz) to the original coordinate X,Y,Z to get the new coordinate X′,Y′,Z′.

J Rotation

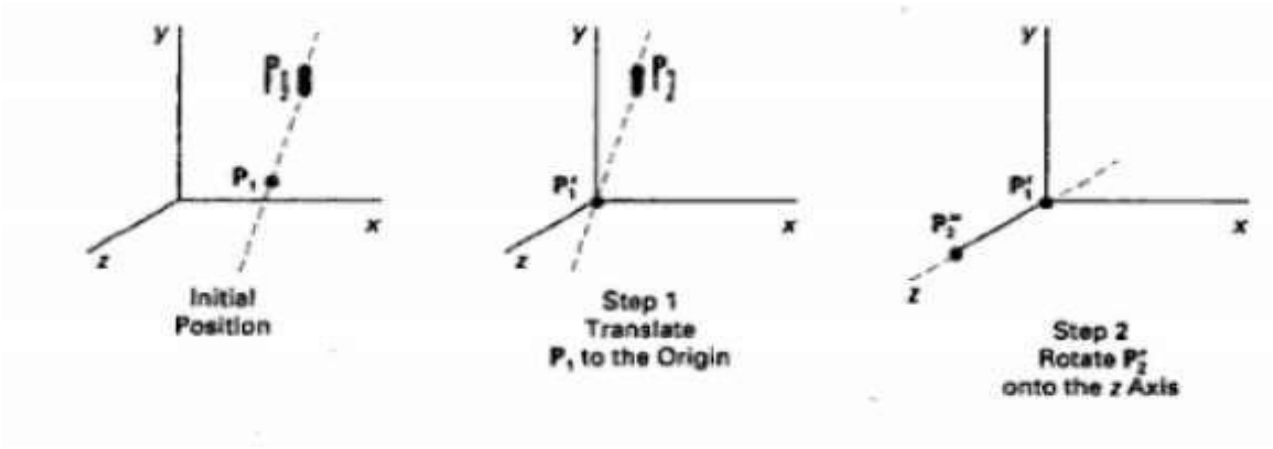
To generate a rotation transformation for an object an axis of rotation must be designed to rotate the object and the amount of angular rotation is also be specified.

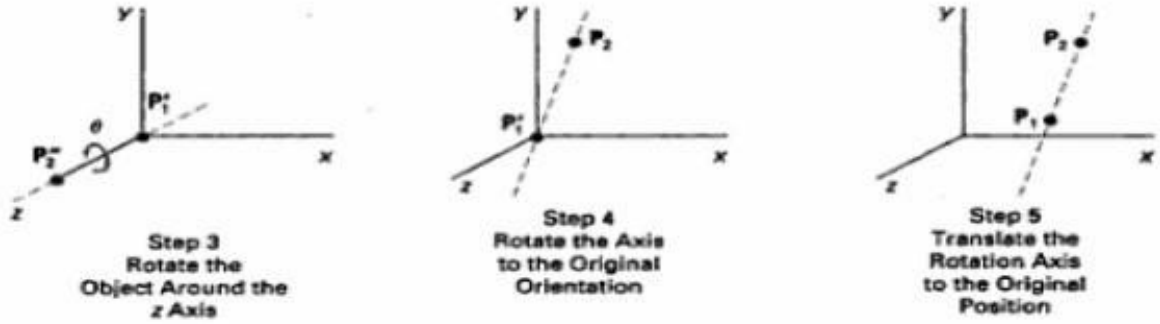
Positive rotation angles produce counter clockwise rotations about a coordinate axis.

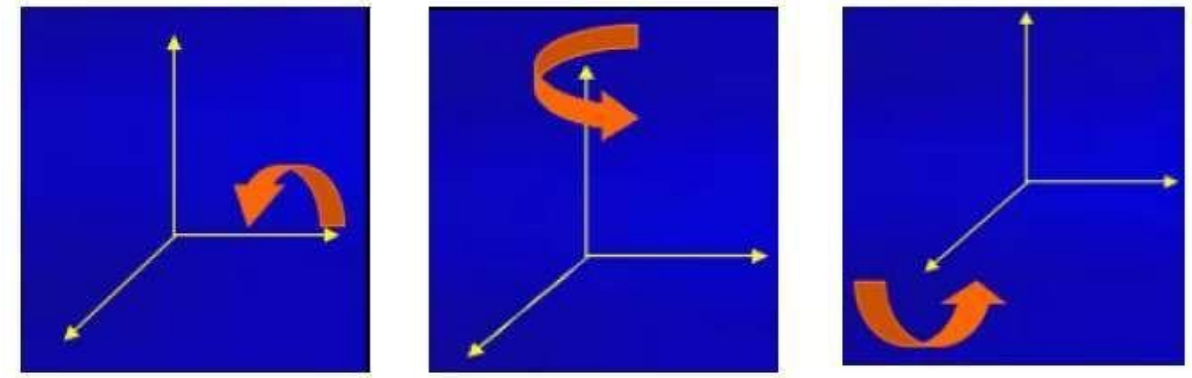
J Co-ordinate Axes Rotations

The 2D z axis rotation equations are easily extended to 3D.

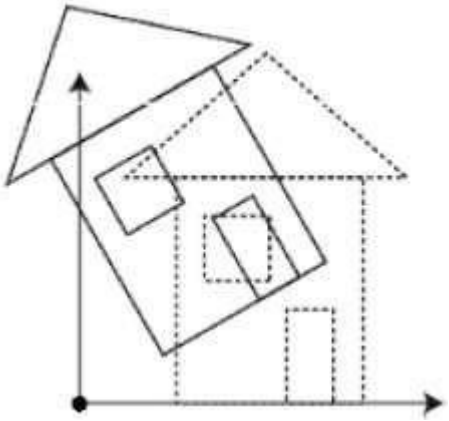
$x = x \cos\theta - y \sin\theta$







Rotation about x-axis Rotation about y-axis Rotation about z-axis



J Scaling

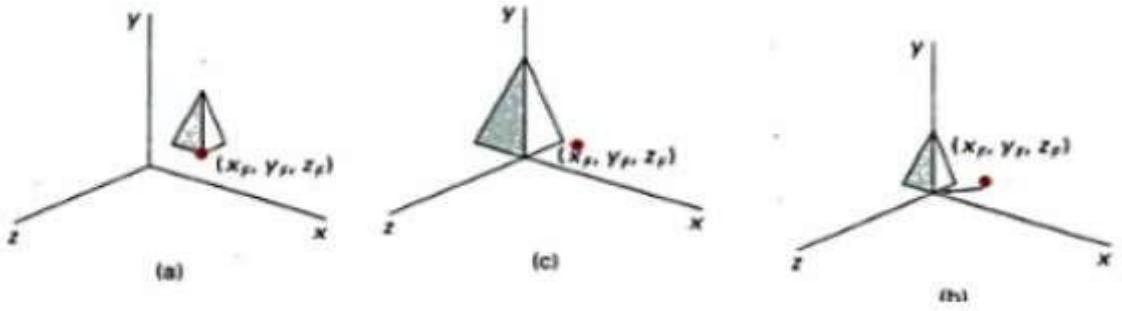
The matrix expression for the scaling transformation of a position P = (x,y,z)

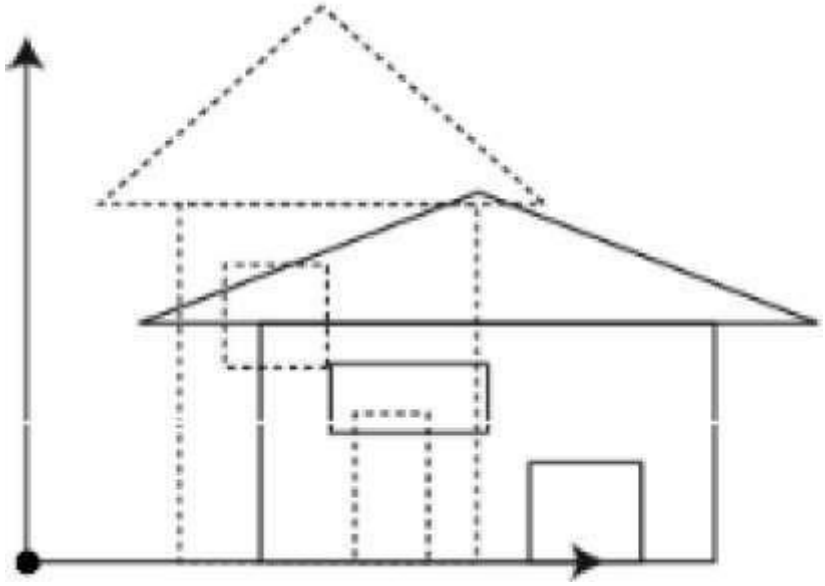
Scaling an object changes the size of the object and repositions the object relatives to the coordinate origin.

If the transformation parameters are not equal, relative dimensions in the object are changed. The original shape of the object is preserved with a uniform scaling (sx = sy= sz) .

Scaling with respect to a selected fixed position (xf, yf, zf) can be represented with the following transformation sequence:

1. Translate the fixed point to the origin.
2. Scale the object relative to the coordinate origin





J 3D Viewing

➢ **Concept:**

- In three-dimensional graphics applications, we can view an object from any spatial position, from the front, from above or from the back.
- We could generate a view of what we could see if we were standing in the middle of a group of objects or inside object, such as a building.

J Viewing Pipeline:

In the view of a three dimensional scene, to take a snapshot we need to do the following steps.

1. Positioning the camera at a particular point in space.
2. Deciding the camera orientation (i.e.,) pointing the camera and rotating it around the line of right to set up the direction for the picture.
3. When snap the shutter, the scene is cropped to the size of the „window of the camera and light from the visible surfaces is projected into the camera film.

In such a way the below figure shows the three dimensional transformation pipeline, from modeling coordinates to final device coordinate.

 Processing Steps

1. Once the scene has been modeled, world coordinates position is converted to viewing coordinates.
2. The viewing coordinates system is used in graphics packages as a reference for specifying the observer viewing position and the position of the projection plane.

3. Projection operations are performed to convert the viewing coordinate description of the scene to coordinate positions on the projection plane, which will then be mapped to the output device.

A **viewplane** or **projection plane** is set-up perpendicular to the viewing Zv axis.

World coordinate positions in the scene are transformed to viewing coordinates, then viewing coordinates are projected to the view plane.

The **view reference point** is a world coordinate position, which is the origin of the viewing coordinate system. It is chosen to be close to or on the surface of some object in a scene.

2. Then we select the positive direction for the viewing Zv axis, and the orientation of the view plane by specifying the **view plane normal vector, N**. Here the world coordinate position establishes the direction for N relative either to the world origin or to the viewing coordinate origin.

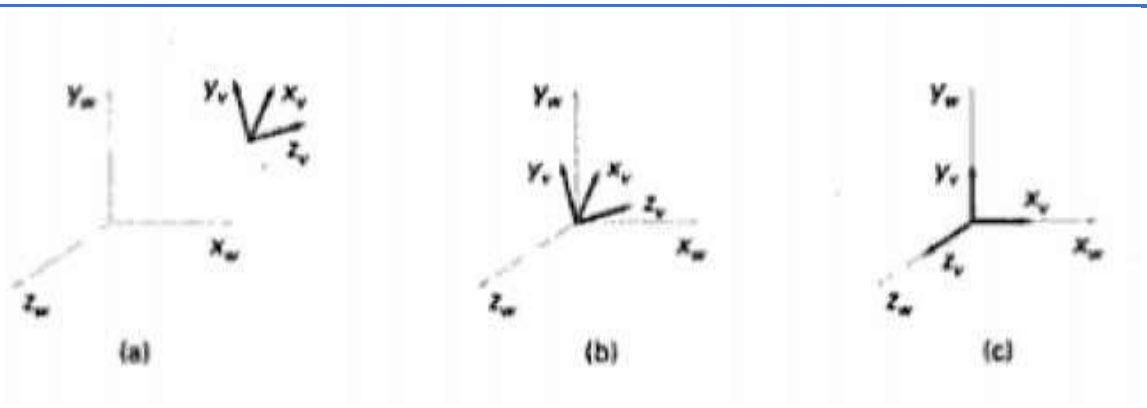


J Transformation from world to viewing coordinates

Before object descriptions can be projected to the view plane, they must be transferred to viewing coordinate. This transformation sequence is,

1. Translate the view reference point to the origin of the world coordinate system.
2. Apply rotations to align the xv, yv and zv axes with the world xw,yw and zw axes respectively.

If the view reference point is specified at world position (x0, y0, z0) this point is translated to the world origin with the matrix transformation.



Another method for generation the rotation transformation matrix is to calculate unit uvn vectors and form the composite rotation matrix directly.

Given vectors N and V, these unit vectors are calculated as

n = N / (|N|)
= (n1, n2, n3) u
= (V*N) / (|V*N|)
= (u1, u2, u3) v
= n*u
= (v1, v2, v3)

This method automatically adjusts the direction for v, so that v is perpendicular to n.

The composite rotation matrix for the viewing transformation is

u1 u2 u3 0 R = v1 v2 v3 0 n1 n2 n3 0 0 0 0 1
which transforms u into the world xw axis, v onto the yw axis and n onto the zw axis

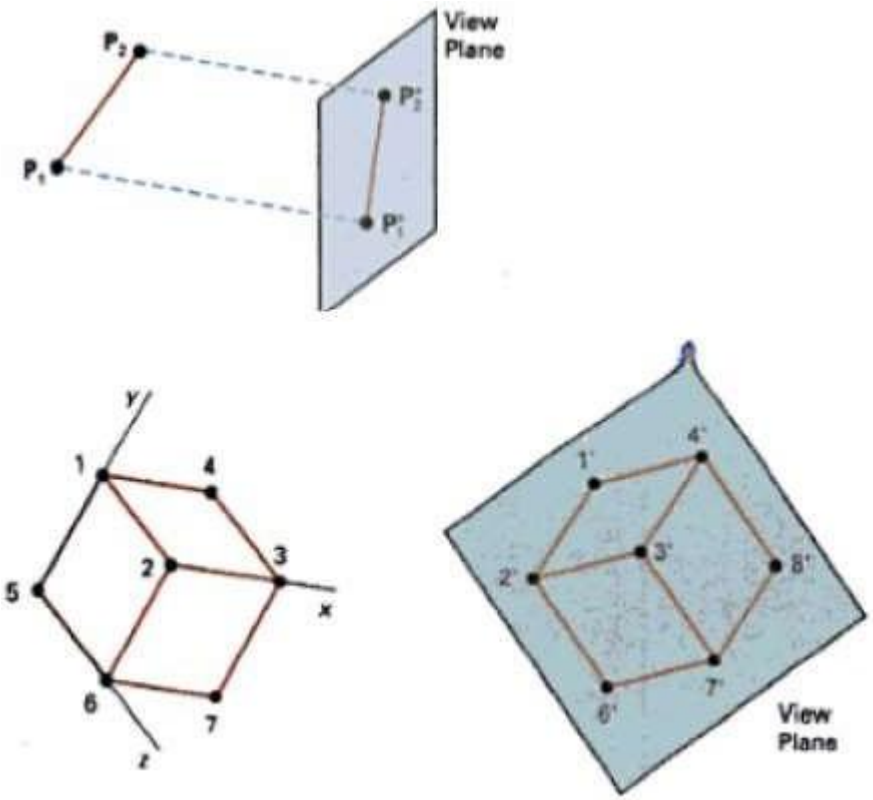
J Projections

Once world coordinate descriptions of the objects are converted to viewing coordinates, we can project the 3 dimensional objects onto the two dimensional view planes.

There are two basic types of projection.

1. **Parallel Projection** - Here the coordinate positions are transformed to the view plane along parallel lines.

Parallel projection of an object to the view plan



J Visible Surface Detection Algorithms

These are classified into two types based on whether they deal with object definitions directly or with their projected images

1. Object Space Methods:

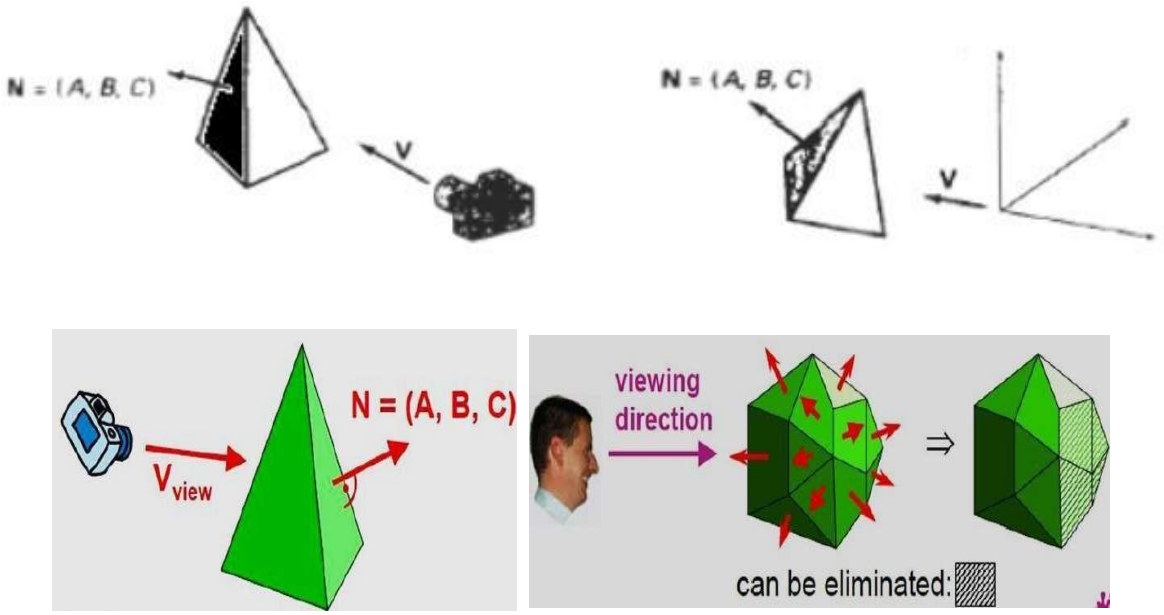
compares objects and parts of objects to each other within the scene definition to determine which surfaces as a whole we should label as visible.

2. Image space methods:

visibility is decided point by point at each pixel position on the projection plane. Most Visible Surface Detection Algorithms use image space methods.

J Back Face Detection

A point (x, y,z) is "inside" a polygon surface with plane parameters A, B, C, and D if Ax + By + Cz + D < 0---------- (1) When an inside point is along the line of sight to the surface, the polygon must be a back face .

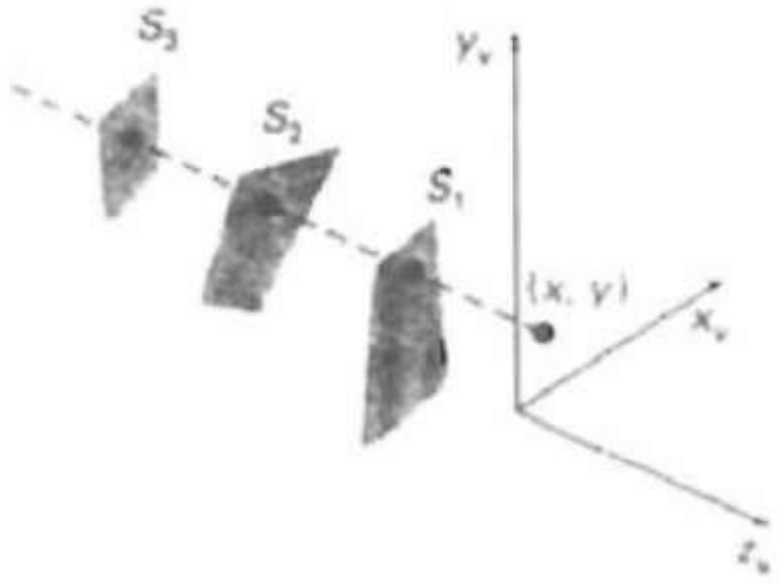


J **Depth Buffer Method / Z- Buffer algorithm:**

A commonly used image-space approach to detecting visible surfaces is the depth-buffer method, which compares surface depths at each pixel position on the projection plane.

This procedure is also referred to as the z-buffer method. Each surface of a scene is processed separately, one point at a time across the surface. The method is usually applied to scenes containing only polygon surfaces, because depth values can be computed very quickly and the method is easy to implement.

But this method can be applied to nonplanar surfaces. With object descriptions converted to projection coordinates, each (x, y, z) position on a polygon surface corresponds to the orthographic projection point (x, y) on the view plane.



We can implement the depth-buffer algorithm in normalized coordinates, so that ***z*** values range from **0** at the back clipping plane to **Zmax** at the front clipping plane.

Two buffer areas are required. A **depth buffer** is used to store depth values for each (x, y) position as surfaces are processed, and the **refresh buffer** stores the intensity values for each position.

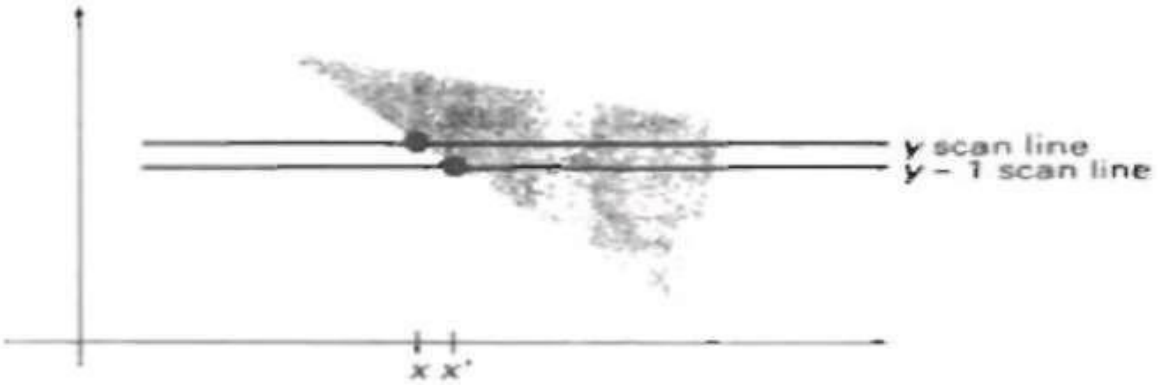
Initially, all positions in the depth buffer are set to 0 (minimum depth), and the refresh buffer is initialized to the background intensity.

We summarize the **steps of a depth-buffer algorithm as follows**:

1. Initialize the depth buffer and refresh buffer so that for all buffer positions (x, y), depth (x, y)=0, refresh(x , y)=Ibackgnd(background color)
2. For each position on each polygon surface, compare depth values to previously stored values in the depth buffer to determine visibility.

Calculate the depth z for each (x, y) position on the polygon.

If z > depth(x, y), then set
depth (x, y)=z , refresh(x,y)= Isurf(x, y)



➢ **Advantages**

- It is easy to implement.
- It reduces the speed problem if implemented in hardware.
- It processes one object at a time.

➢ **Disadvantages**

- It requires large memory.
- It is time consuming process.

J **Illumination models and Surface Rendering**

Illumination model, also known as Shading model or Lightning model, is used to calculate the intensity of light that is reflected at a given point on surface. There are three factors on which lightning effect depends on:

1. **Light Source :**
   Light source is the light emitting source. There are three types of light sources:
   1. **Point Sources –** The source that emit rays in all directions (A bulb in a room).
   2. **Parallel Sources –** Can be considered as a point source which is far from the surface (The sun).
   3. **Distributed Sources –** Rays originate from a finite area (A tubelight).

   Their position, electromagnetic spectrum and shape determine the lightning effect.
2. **Surface :**
   When light falls on a surface part of it is reflected and part of it is absorbed. Now the surface structure decides the amount of reflection and absorption of light. The position of the surface and positions of all the nearby surfaces also determine the lightning effect.
3. **Observer :**
   The observer's position and sensor spectrum sensitivities also affect the lightning effect.

Added realism is attained in displays by setting the surface intensity of objects according to the lighting conditions in the scene and according to assigned surface characteristics. Lighting specifications include the intensity and positions of light sources and the general background illumination required for a scene. Surface properties of objects include degree of transparency and how rough or smooth the surfaces are to be. Procedures can then be applied to generate the correct illumination and shadow regions for the scene.

- **Exploded and Cutaway View**

Exploded and cutaway views of such objects can then be used to show the internal structure and relationship of the object Parts

- **Three-Dimensional and Stereoscopic View**

Three-dimensional views can be obtained by reflecting a raster image from a vibrating flexible mirror. The vibrations of the mirror are synchronized with the display of the scene on the CRT. As the mirror vibrates, the focal length varies so that each point in the scene is projected to a position corresponding to its depth.

Stereoscopic devices present two views of a scene: one for the left eye and the other for the right eye.